

**“I can resist
anything except
temptation”**

Oscar Wilde's 1892
Lady Windermere's Fan



Playful Programming and Artificial Intelligence

Course Overview & Intended Learning Outcomes

24 August 2026

What this workshop is

A workshop for high school teachers who are curious about AI and want space to explore it seriously, on their own terms.

Not a fixed syllabus of AI facts. AI is treated as something worth **arguing about, building with, and getting wrong** in productive ways.

No prior AI background assumed. The aim isn't finished materials — it's a set of ideas, questions, and approaches participants keep using after the course ends.

The organising idea: didactic transposition

Moving from scholarly knowledge to teachable content isn't simplification alone — it's careful **reconstruction**.

Participants first meet AI concepts as learners — through physical activity, hands-on tinkering, and deliberate confusion — then do the harder work: *what would this need to become, to make sense in my own classroom?*

How it's structured

Block	Dates	Focus
August Day 1	Mon 24 Aug	Physical AI: sensing, classification, unplugged
August Day 2	Tue 25 Aug	Generative AI: LLMs, prompting, sensor + language
August Day 3	Weds/Thu 26–27 Aug	Project: design, build, and present
October Day 4	Mon 5 Oct	Classroom feedback: what happened, what changed
October Day 5	Tue 6 Oct	Consolidation: refine, share, and publish

Between blocks, participants try an idea with their own peers and/or students.

Intended Learning Outcomes

By the end of the workshop, participants will be able to:

Understand & evaluate

- **Interrogate what counts as "AI"** in everyday objects and systems, and articulate that reasoning to others
- **Explain didactic transposition** and use it to move scholarly AI knowledge toward something genuinely teachable
- **Describe, conceptually, how large language models work** and name their common failure modes — hallucination, bias, opacity
- **Judge what makes a prompt or instruction effective**, and where the analogy to good pedagogy holds or breaks

Build, design & reflect

- **Build simple browser-based sensing/classification prototypes** (p5.js + ml5.js + Claude) with zero required programming, and explain the sensor → classifier → language-model pipeline underneath
- **Design a classroom-ready IoT + GenAI activity** of their own — cheap, safe to fail, genuinely fun — and playtest it with peers
- **Try the activity with real students**, then revise it honestly based on what actually happened
- **Take a position on responsible AI literacy** at gymnasium level: what to show, what to hold back, and why

Where to go next

Full programme: [Didactic Workshop IOT details.md](#)

Prototypes: [p5.js collection](#)

The workshop doesn't resolve how to teach AI well. It tries to make that question more interesting, more concrete, and more worth pursuing.